



■ THE SAFETY LAYER FOR ADDITIVE MANUFACTURING

Firearm screening for 3D printing.

Identify firearm-related geometry before a file is published, queued or printed. PrintGuard brings local screening and auditable decisions to your existing workflow.

Tested on printers | Runs on your infrastructure | Your files stay with you

BENCHMARK PERFORMANCE

PRINTGUARD V1 · JULY 2026

96.75%

Firearm detection

0.06%

False-positive rate

<1 second

Per-file scoring

10,000

Source-separated test files

~10x

Growth in reported 3D-printed gun recoveries in a U.S. police-department sample, 2020–2024.¹

A growing risk. An international need.

Downloadable designs and accessible printers move firearm production beyond traditional supply chains. Europol has documented rising seizures of 3D-printed firearms across Europe.²

Platforms and printer operators need to identify risky files while keeping legitimate design and manufacturing moving.

Screen the shape. Retain the evidence.

From model intake to policy action

01 / INTAKE

Parse the geometry

Read and validate the mesh.



02 / ANALYSIS

Measure & classify

Extract features and score risk.



03 / ACTION

Apply your policy

Continue, review or hold.

Machine-learning classifiers analyze mesh statistics, point clouds, voxel occupancy and multiview projections to assess risk, part type and firearm family. Exact SHA-256 and near-geometry matching add known-file evidence. Uncertain results can be routed for review.

Built for the places files flow.

Printer makers & slicers: screen model imports before slicing or print submission.

Marketplaces & file platforms: add geometry-based evidence to upload and publication review.

Print services & shared fleets: screen customer jobs, classroom files and lab intake.

Local control. Practical integration.

Deploy locally. Run an offline Linux service through a CLI, Python or local HTTP interface, without centralizing customer geometry.

Keep an audit trail. Record hashes, scores, match evidence and policy actions so decisions can be reviewed and reproduced.

Supported mesh formats STL · OBJ · AMF · 3MF · OFF / STEP and STP require conversion to a mesh.

See PrintGuard in action.

Book a private demo and explore an integration with your team.

contact@printguard.cv ↗

Benchmark: Detection is the share of firearm files flagged; false positives are the share of benign files flagged. Timing is for average-size files and varies with hardware and file complexity. Screening informs policy decisions; it is not a legal determination.

Sources: 1. AP, Oct. 16, 2025, reporting Everytown data from about two dozen U.S. police departments; not a national estimate. 2. Europol, May 27, 2022.

Local screening. Auditable decisions.

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